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The Mathematics and Design of WaterStrider

Technical Report

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Abstract

WaterStrider is a per-sensor statistical detector for Strawberry Creek. It removes each sensor's normal daily rhythm, runs three complementary tests on what remains (a rolling Hampel test for spikes, a two-sided cumulative sum test for shifts that hold, and an exponentially weighted moving average test for slow drift), and sets every threshold from a tail fitted to clean data at a false alarm rate chosen in advance. Nothing in the model is trained by gradient descent. Every number in it is a median, a deviation or a fitted tail, so each alert can be explained in one sentence. WaterStrider meets the same detector contract as DuskCrayfish and CobbleShoal, and it also serves as the simple baseline that the graph models have to beat on the same labelled events. Here, we set out the mathematics of the residuals, the three tests and the calibration, the departures from the original plan that the data required, and what the shipped calibration measured. In brief, the calibration holds its false alarm budget on held-out months, but at that budget WaterStrider caught none of the four labelled anomalies.

1 Introduction

The starting point for WaterStrider was Hampel's X84 rule, which flags a value more than k median absolute deviations (MAD) from the median, with $k = 5.2$ in Hampel's original suggestion [Hampel, 1985]. The median and the MAD are not pulled around by the outliers they are looking for, the rule needs no training, and its alerts explain themselves. On this creek, however, the rule alone is a poor detector, for five reasons. (1) A shift that lasts becomes the new median after half a window, so only its start is flagged, while a discharge can hold a new level for hours. (2) A slow drift, such as probe fouling, never stands out from its own window. (3) Storms and daily cycles look unusual, since rain dilutes conductivity and water temperature swings every day. (4) A flat signal, such as depth at base flow, has a MAD near zero, so every small change looks extreme. (5) A fixed k ignores how many tests run. The plan counted 17 sensors reporting 96 times a day, about 1,632 tests a day per rule. Work on automated anomaly detection in river turbidity, conductivity and level points the same way, and recommends combining methods to catch more events while keeping false detections low [Leigh et al., 2019].

WaterStrider keeps the Hampel test and adds what each of these limits calls for: a cumulative sum (CUSUM) test for shifts that hold, an exponentially weighted moving average (EWMA) test for slow drift, a daily profile and the rain gate for daily cycles and storms, a floor on every spread for flat sensors, and a calibrated threshold for each test in place of a fixed k. Each test covers a failure the other two miss. An interquartile range fence or a plain z-score on the same window would be close to the same test as Hampel's, and would add false alarms without catching anything new.

Table 1: The calibrated configuration. These are the settings the shipped calibration file records, not defaults or recommendations.

Setting	Value
Sampling grid	15 min, 96 steps a day
Level window	12 hours (48 steps), trailing median
Daily profile	median over the previous 14 days
Share of real readings a window needs	0.75
MAD scale	1.4826
Spread floor	larger of probe resolution and 0.10× spread
Hampel window w	13 steps, including the current one
CUSUM allowance k	0.5
EWMA weight λ	0.2
Gap that resets CUSUM and EWMA	more than 8 missing steps
Declustering gap	8 steps
Tail start percentiles	98, then 95, 90, 85, 80
Minimum clusters for a tail	20
Model false alarm rate q	10^{-4}
Rate per test $q/3$	3.33×10^{-5}
Calibration span	4 June 2025 to 29 April 2026
Split	even months fitted, odd months held out
Sensors calibrated	30 (10 sites × 3 variables)

In StrawberryWatch, a model is whatever meets the detector contract, and the contract does not ask for weights. A detector built from medians and fitted tails is therefore as much a model as DuskCrayfish. WaterStrider also does a second job. Any graph neural network invites the comparison with a simple method on the same data, and WaterStrider provides that comparison on the same labelled events and at the same false alarm budget, which is a fairer comparison than X84 with a guessed k.

1.1 The calibrated configuration

Table 1 lists the settings the shipped calibration was fitted at, read from the model definition and from `water_strider_calibration.json`, which was made on 14 October 2026. We refer to these values throughout the mathematics that follows.

2 Data and Notation

2.1 Readings on the grid

Each sensor is judged on its own readings. We floor every timestamp to the quarter hour and average the readings that land on the same grid point, so a sensor becomes one value per 15 min step, with a gap where nothing arrived. We never fill gaps. DuskCrayfish showed that a filled-in newest reading causes false alarms, so instead each window is judged only when at least three quarters of its readings are real. For calibration, we drop readings outside their plausible range and readings from before the probe's recorded install date, since those are not that probe's own.

2.2 Level, daily profile and residuals

The tests run on residuals, the reading with its normal level and daily rhythm removed. For sensor readings x_t, the level L_t is the median of the 48 readings before step t, which is 12 hours of history. The daily profile $P(\cdot)$ assigns a value to each of the 96 quarter hours of the Pacific day. For each calendar day we take the median of $x_t - L_t$ at each quarter hour, and the profile in force on a day is the median of those daily values over the previous 14 days. A quarter hour seen on fewer than three of those days has no rhythm to remove yet, and its profile value is zero. The residual is then

$$r_t = x_t - L_t - P\big(\mathrm{slot}(t)\big), \tag{1}$$

so a normal afternoon temperature peak scores close to zero. Medians keep a past event from bending the profile, and the whole computation costs almost nothing.

Table 2: Notation.

Symbol	Meaning	Dimension
x_t	one sensor's reading at step t	scalar
L_t	level, 12 hour trailing median	scalar
$P(\cdot)$	daily profile by quarter hour	96
r_t	residual	scalar
s	sensor spread, scaled MAD	scalar
f	spread floor	scalar
z_t	standardised residual, $r_t / \max(s, f)$	scalar
H_t, C_t, E_t	Hampel, CUSUM and EWMA statistics	scalar
ρ	lag-one autocorrelation of the residuals	scalar
t_0	tail start, a percentile of the clean statistic	scalar
$\hat{\sigma}, \hat{\gamma}$	fitted tail scale and shape	scalar
n	clean fitted steps	scalar
N_t	clusters above t_0	scalar
q	false alarm rate for the whole model	scalar, 10^{-4}
z_q	a test's threshold at rate $q/3$	scalar

The level is a departure from the plan, which subtracted the daily profile alone. We tried level windows of 6 and 24 hours as well as 12. At 24 hours the shift test went over budget on the held-out months, and at 6 hours a drift was absorbed into the level too quickly to be seen.

2.3 Spread and its floor

Each sensor has one robust spread, the scaled MAD of its residuals over the clean fitted steps,

$$s = 1.4826 \cdot \text{median} \left| r_t - \text{median}(r) \right|, \tag{2}$$

where the factor makes s approximately one standard deviation for normal data. Every spread in WaterStrider is a scaled MAD, and the code names say so. The distinction matters, because 5.2 raw MADs is about 3.5 standard deviations, while 5.2 scaled MADs is about 5.2.

A steady sensor's MAD can be close to zero, which makes every small change look extreme. We therefore floor every spread at

$$f = \max\left(\varepsilon_v,\ 0.10\, s\right), \tag{3}$$

where ε_v is the smallest step the probe reports: 1.0 for conductivity and depth, 0.1 for temperature and floating conductivity, and 0.01 for dissolved oxygen, each in that probe's reporting units. Table 2 collects the symbols used in the rest of this document.

3 The Three Tests

3.1 Rolling Hampel test

The Hampel statistic is the distance of a residual from the median of its window, in floored scaled MADs,

$$H_t = \frac{\left| r_t - \text{median}(\mathcal{W}_t) \right|}{\max\left(1.4826 \cdot \text{MAD}(\mathcal{W}_t),\ f\right)}, \tag{4}$$

where $\mathcal{W}_t$ holds the $w = 13$ residuals ending at step t, about three hours. The window is long enough that one bad reading cannot drag the median, and short enough that a lasting shift has been absorbed by the time the CUSUM test has it. The window includes the step itself, so a spike is judged against the readings around it. This is the moving-window form of the rule known as the Hampel identifier. The test catches single spikes and sudden drops, and it misses shifts that last past half the window.

3.2 Two-sided CUSUM

The CUSUM test accumulates standardised residuals beyond a small allowance, in both directions [Page, 1954],

$$C_t^+ = \max\left(0,\ C_{t-1}^+ + z_t - k\right), \qquad C_t^- = \max\left(0,\ C_{t-1}^- - z_t - k\right), \qquad C_t = \max\left(C_t^+, C_t^-\right), \tag{5}$$

with $k = 0.5$, the usual choice for catching a shift of about one spread. A shift that holds keeps adding to one of the sums, while a single spike adds once and decays. The test therefore catches shifts that hold and misses single spikes by design. Missing steps are skipped, and both sums reset after more than 8 consecutive missing steps, so a long gap does not carry a sum across it.

We also tried scaling the CUSUM input for slow wander. Residuals here are strongly autocorrelated, and a running sum of autocorrelated values spreads wider than one of independent values by the long-run factor $\sqrt{(1+\rho)/(1-\rho)}$. The scaling shrank real shifts below the allowance, and the shift test then barely fired, so the shipped calibration has it turned off.

3.3 EWMA

The EWMA test averages the standardised residuals with geometrically decreasing weights and reads the average in units of its own spread,

$$\bar{z}_t = \lambda z_t + (1-\lambda)\,\bar{z}_{t-1}, \qquad E_t = \frac{|\bar{z}_t|}{\sqrt{\lambda/(2-\lambda)}}, \tag{6}$$

with $\lambda = 0.2$, within the range of 0.2 to 0.3 that the NIST handbook suggests [NIST/SEMATECH]. Averaging narrows the spread by $\sqrt{\lambda/(2-\lambda)}$, which is why the limits of an EWMA chart are narrower than those of the raw data. The test catches slow drift and misses sudden changes. It resets after long gaps in the same way as CUSUM.

3.4 Combining the tests

A step is flagged when any test crosses its own threshold, and the detector returns one finding per site, sensor and test. CUSUM and EWMA build up over time, so their statistics already describe changes that last. A Hampel spike, by contrast, counts only when a second reading agrees, either the step before or another sensor at the same site. Writing $R_t = H_t/z_q$ for the spike's share of its own threshold and R'_t for the largest share among the site's other sensors at the same step, the Hampel score that is compared with the threshold is

$$\tilde{H}_t = z_q \cdot \min\big(R_t,\ \max(R_{t-1}, R'_t)\big). \tag{7}$$

A single bad reading is the job of the Settling Pool screen described in the support modules report, and WaterStrider does not repeat its checks.

4 Calibration

4.1 Clean steps and the split

We computed every statistic over the full calibration span, with CUSUM and EWMA carrying their sums as they would in a live check, and fitted only on clean steps. A step is clean when it lies outside all eight labelled event spans and the rain gate would not be raising the threshold there. Each sensor needed at least 14 clean days in the fitted months to be calibrated at all.

We fitted on even months and held out odd months, so that both sides contain wet and dry weather. The calibration script records that a chronological split put the rain on the held-out side. Measured over the span, the share of steps where the rain gate raises the bar is 21.0% in even months and 10.9% in odd months, against 14.1% and 22.5% for a chronological split at 1 February 2026, so both splits leave wet weather on both sides.

4.2 Declustered tails

The residuals are slowly varying. Their lag-one autocorrelation has a median of 0.963 across the 30 calibrated sensors, with a range of 0.701 to 0.993, and CUSUM and EWMA stay over a line for many steps once they cross it. Steps above a threshold are therefore far from independent, and a tail fitted to every exceedance would describe the length of a few runs rather than how often a run begins. We declustered the statistic, which is the usual remedy when exceedances are not independent [Coles, 2001]. Runs over the starting threshold that lie within 8 steps of each other count as one cluster, and we fitted the tail to the cluster peaks. In this form N_t counts clusters and q is a rate of alarms per step, not of steps over the line.

Table 3: The 90 fitted tails by test. A pooled tail is shared by every sensor of one kind. The median threshold is in the units of each statistic.

Test	Own	Pooled	From 98th	$\hat{\gamma} < 0$	$\hat{\gamma} > 1$	Median z_q
Hampel	30	0	28	2	3	38.5
CUSUM	5	25	0	3	10	3,838.0
EWMA	29	1	1	5	5	113.0

For each sensor and test, the tail start t_0 is the 98th percentile of the clean fitted statistic, as in the peaks-over-threshold method of Siffer et al. [2017]. CUSUM and EWMA can spend the top 2% of steps in a few long runs, so where fewer than 20 clusters exceed t_0 we step down to the 95th, 90th, 85th and 80th percentiles in turn. A Generalised Pareto distribution with location zero is then fitted by maximum likelihood to the cluster peaks minus t_0 [Pickands III, 1975], and the threshold for a test's share of the budget is

$$z_q = t_0 + \frac{\hat{\sigma}}{\hat{\gamma}} \left[\left(\frac{(q/3)\, n}{N_t} \right)^{-\hat{\gamma}} - 1 \right], \qquad z_q = t_0 - \hat{\sigma} \log\!\left(\frac{(q/3)\, n}{N_t} \right) \quad \text{as } \hat{\gamma} \to 0, \tag{8}$$

using the exponential limit whenever $|\hat{\gamma}| < 10^{-6}$.

A sensor with fewer than 20 clusters even from the 80th percentile shares a tail with every sensor of its kind. We place the kind's statistics end to end, with a gap between sensors that no cluster can cross, and fit one tail to the pooled peaks. The plan fitted a tail per sensor; pooling is a departure that the thin sensors required.

4.3 The budget

The model's false alarm rate is $q = 10^{-4}$, the same default the support modules use, split evenly across the three tests so that each test gets $q/3 = 3.33 \times 10^{-5}$. Splitting by addition is a union bound, so the real rate of the three tests together is at most q [Dunn, 1961], and because the tests share residuals it is likely lower.

The calibration file keeps the whole fit for every sensor and test (t_0, $\hat{\sigma}$, $\hat{\gamma}$, n and N_t) as well as the threshold, together with q, the span and split used, the settings and a hash of the fitted data. Keeping the fit, and not only the threshold, is what lets the sensitivity setting in Section 5.1 move the rate without recalibrating.

4.4 The fitted tails

We calibrated 30 sensors, conductivity, depth and temperature at each of 10 sites, and skipped 11. Nine of the skipped sensors had too few readings, including every sensor at scnf010 and the dissolved oxygen and floating conductivity probes at botanical_garden and south_fork_0. The remaining two, dissolved oxygen and floating conductivity at south_fork_3, had only 10 and 12 EWMA clusters even from the 80th percentile, and no pooled tail could be fitted for them. The calibrated sensors had between 16.5 and 88.9 clean days in the fitted months. Table 3 summarises the 90 tails.

The CUSUM tails depend most on pooling. All 10 conductivity sensors and all 10 depth sensors share one CUSUM tail each, as do 5 of the 10 temperature sensors, and only one EWMA tail is pooled. The pooled conductivity CUSUM threshold is 3,838.0 and the pooled depth CUSUM threshold is 221,574.1, because the depth tail's fitted shape is $\hat{\gamma} = 2.77$. A large positive shape places the threshold far beyond the observed peaks. The same holds for a few sensors' own tails, such as the kingman_hall conductivity EWMA ($\hat{\gamma} = 2.00$, $z_q = 73{,}312.4$) and the oxford depth Hampel test ($\hat{\gamma} = 2.83$, $z_q = 65{,}418.9$).

4.5 The held-out check

A calibration passes only if the alarm rate on the held-out clean steps comes within twice the budget. Table 4 gives the result. Every test passed, and so did the three tests together.

Table 4: Alarms on held-out clean steps (odd months). An alarm is a run over the threshold, with runs closer than 8 steps counted as one. The allowance is twice the budget.

Test	Alarms	Steps	Rate	Allowed
Hampel	5	172,713	2.89×10^{-5}	6.67×10^{-5}
CUSUM	5	173,961	2.87×10^{-5}	6.67×10^{-5}
EWMA	2	173,961	1.15×10^{-5}	6.67×10^{-5}
Any test	9	173,961	5.17×10^{-5}	2.00×10^{-4}

Table 5: The sensitivity stops, as the model's overall rate. Each test gets a third. The expected flags assume 1,632 tests a day per rule, from the plan's count of 17 sensors.

Stop	Rate q	Expected false flags
Far fewer	10^{-5}	about one in 2 months
Fewer	3×10^{-5}	about one in 3 weeks
Calibrated	10^{-4}	about one a week
More	3×10^{-4}	about one every 2 days
Far more	10^{-3}	about one and a half a day

5 Serving

5.1 Manual sensitivity

The calibration sets the default, and a sensitivity setting lets people move away from it in steps that stay statistically sound. The setting moves the false alarm rate rather than the thresholds. Each sensor's tail has its own shape, so scaling every threshold by the same factor would make one sensor much noisier and barely change another, whereas moving the rate keeps every sensor equally strict. Each test can be set on its own, and every new threshold comes from the tail already fitted, so nothing is refitted. Table 5 lists the five stops.

The range stops there for two reasons. Above about 0.02 per test, the threshold would fall below the 98th percentile that the tail fit starts from, and the method no longer applies. Well below 10^{-5}, the threshold lands far beyond anything in the calibration data, and the number becomes a guess.

A hand-typed threshold z for one sensor and test runs the threshold formula backwards, to give the rate it implies,

$$q = \frac{N_t}{n}\left(1 + \frac{\hat{\gamma}\,(z - t_0)}{\hat{\sigma}}\right)^{-1/\hat{\gamma}}, \qquad q = \frac{N_t}{n}\, e^{-(z-t_0)/\hat{\sigma}} \ \text{ as } \hat{\gamma} \to 0, \tag{9}$$

and a typed threshold past a negative shape's finite endpoint implies a rate of zero. At the calibrated stop, the stored threshold is used unchanged.

5.2 Rain

We calibrated on dry-weather steps, and during and after rain the rain gate raises each threshold by a multiplier, as it does for the other models. The gate's default rule doubles the bar while it rains and then returns it to the base value linearly over 36 hours, counting a reading above 0.1 mm as wet. WaterStrider declares rain as built-in support behaviour, so attaching a separate rain modulator is refused, since it would raise the thresholds twice.

5.3 The detector contract

WaterStrider reads one JSON file, the calibration, which the existing file check verifies by hash, and it imports no deep learning library. A check is medians and running sums over short windows for each sensor, which takes milliseconds and a few megabytes. Live checks read two days of readings, which is not enough to rebuild a 14 day profile, so the served detector uses the latest profile stored in the calibration as a fixed shape and computes the level from the live readings. Table 6 gives its answers to the rest of the contract.

Table 6: WaterStrider's answers to the detector contract.

Contract piece	WaterStrider
Sites	in-service sites with a calibrated sensor
Rules	`hampel_spike`, `cusum_shift`, `ewma_drift`
Plain wording	a sudden spike, a jump that stayed, a slow drift
Findings	one per site, sensor and rule over its rain-gated line
Steps over the line needed	1
Site choices	don't flag, and leave out
Built-in support	rain
Weather	needed while the rain gate is on
Files	`water_strider_calibration.json`

Table 7: False flags per sensor-week on held-out clean data, network wide. X84 was counted on conductivity sensors only.

Method	Flags per sensor-week
WaterStrider	0.039
Hampel alone	0.019
CUSUM alone	0.023
EWMA alone	0.008
Naive X84, $k = 5.2$	5.308
Naive X84, $k = 347.3$	0.409

6 Evaluation

We compared WaterStrider with each of its three tests alone, with naive X84 (a fixed 5.2 raw MADs from a trailing median, with no daily profile and no rain handling), and with DuskCrayfish and CobbleShoal, each with its shipped checkpoint and default settings. We count whole events rather than single readings. We ran the evaluation on 25 October 2026, against the calibration described above.

6.1 False flags on held-out months

Table 7 gives the false flags per sensor-week on the clean steps of the held-out months, over 258.9 sensor-weeks. The budget allows about 0.07 a sensor-week. WaterStrider ran at 0.039, about 1.2 a week across the 30 sensors.

Naive X84 at $k = 5.2$ flagged about 136 times as often as WaterStrider. The evaluation then searched for the k at which X84 matches WaterStrider's rate, up to $k = 2{,}000$, and the closest it came was 0.409 a sensor-week at $k = 347.3$, still about ten times WaterStrider's rate. X84 cannot reach that rate at any k, because on a flat stretch the raw MAD is zero and any change at all exceeds every multiple of it.

6.2 Labelled events

Table 8 gives which methods flagged conductivity at the labelled site in each graded event. The four labelled anomalies are the June 2025 discharge on the south fork, both halves of the September 2025 overnight event on the south fork, and the March 2026 hydrant break into the north fork.

WaterStrider caught none of the four, and neither did any of its tests alone. Naive X84 caught all four at $k = 5.2$, at 136 times the false flag rate, and one at $k = 347.3$. In the same run, DuskCrayfish flagged three of the four (the June discharge and the September event at south_fork_1, and the hydrant break) and CobbleShoal flagged none.

The events that are not graded give a similar picture. WaterStrider flagged the November 2025 foam event at north_fork_0 through its Hampel test, 40.5 hours after the earliest flag from any method, which was X84's. It did not flag the June 2025 discharge at south_fork_2 or the August 2025 sprinkler runoff, and no method flagged the botanical garden actuator event. In both rain-only events, the November 2025 rain and the April 2026 rainfall, WaterStrider flagged conductivity at one site, codornices, where X84 flagged 7 and 8 sites and DuskCrayfish 3 and 1.

Table 8: Labelled anomalies flagged at the labelled site's conductivity sensor. A tick means the method flagged the event; X84 matched is X84 at $k = 347.3$.

Event	Site	WS	Hampel	CUSUM	EWMA	X84	X84 matched
June 2025 discharge	south_fork_1					✓	
September 2025 overnight	south_fork_1					✓	
September 2025 overnight	south_fork_2					✓	✓
March 2026 hydrant	north_fork_0					✓	
Caught		0	0	0	0	4	1

Table 9: The largest share of its rain-gated line each test reached at the labelled site's conductivity sensor, at the calibrated rate.

Event	Site	Hampel	CUSUM	EWMA
June 2025 discharge	south_fork_1	0.062	0.172	0.081
September 2025 overnight	south_fork_1	0.937	0.145	0.148
September 2025 overnight	south_fork_2	0.098	0.164	0.105
March 2026 hydrant	north_fork_0	0.256	0.099	0.237

6.3 Peaks against thresholds

Table 9 shows how close each test came. For every step of the event window we divided the test's score by its rain-gated line, and the table gives the largest share each test reached. A value of 1 would be a flag.

Only one event came close. The Hampel test at south_fork_1 in the September event reached 0.937 of its line. Every CUSUM peak stayed below a fifth of the pooled conductivity threshold of 3,838.0, so the shift test, the one meant for discharges that hold a new level, was furthest from firing.

6.4 Sensitivity stops

We replayed the same events at each sensitivity stop. Table 10 gives the result.

At the More stop, the Hampel test caught the September event at south_fork_1. At the Far more stop, the EWMA test also caught the hydrant break, but the rain-only events then flagged 4 and 6 sites. The flags at codornices in the rain-only events did not go away at any stop, since their peaks were several times the line.

6.5 Pass criteria

The plan set four pass criteria, and the shipped calibration meets one of them. (1) False flags on held-out clean data came within twice the budget, which passed. (2) It did not catch the labelled discharge events. (3) It flagged one site in each of the labelled rain-only events. (4) At the same false flag rate as naive X84, it caught 0 events against X84's 1.

7 Limitations and Future Work

The main limitation is that the calibrated thresholds are too far above the events. WaterStrider holds its false alarm budget on held-out months, but at that budget it catches none of the four labelled anomalies, and at the loosest stop it catches two while flagging most sites in a rain-only event. As a baseline, it therefore does not yet set a meaningful bar for the graph models. That said, DuskCrayfish flagged three of the same four events in this evaluation, so the events are detectable from these sensors.

The CUSUM tails are the clearest place to look. Most of them are pooled, the pooled conductivity threshold sat 5.8 to 10.1 times above the four events' peaks, and the depth tail's large fitted shape put its threshold at 221,574.1. A Generalised Pareto tail with $\hat{\gamma} \geq 1$ has no finite mean, and a threshold extrapolated from such a fit may say more about the fit than about the creek. We recommend checking the tail fits against the held-out peaks, and trying a clamped shape, as CobbleShoal clamps γ at zero for its p-values, before tuning anything else.

Table 10: Labelled anomalies caught, and sites whose conductivity was flagged in the November 2025 and April 2026 rain-only events, at each stop.

Stop	Rate q	Anomalies caught	Rain-only sites flagged
Far fewer	10^{-5}	0 of 4	1 and 1
Fewer	3×10^{-5}	0 of 4	1 and 1
Calibrated	10^{-4}	0 of 4	1 and 1
More	3×10^{-4}	1 of 4	1 and 1
Far more	10^{-3}	2 of 4	4 and 6

The level window may remove part of what the tests are looking for. A 12 hour trailing median takes on a new level once that level has held for 6 hours, half the window, so the residual of a discharge that holds longer returns toward zero. This may be one reason the shift test stays quiet. We chose 12 hours over 6 and 24 on the false alarm budget alone, and no event was part of that choice.

The spike test needs a second reading to agree. A real one-step spike in conductivity is therefore counted only when the depth or temperature sensor at the same site spikes at the same step, and otherwise not at all. Settling Pool, which would take over the job of rejecting single bad readings and so allow the corroboration rule to be relaxed, is not yet built.

The reason recorded for the blocked split does not hold on the counts in Section 4.1. Both splits leave rain on both sides, and the blocked split puts about twice as many gated steps in the fitted months as in the held-out ones. The split is still a reasonable one, but it was not chosen for the reason the calibration script states.

The served detector uses a fixed daily profile from the end of the calibration span. The plan rebuilt a sensor's profile and thresholds after a replacement, using the install dates in the inventory. The calibration drops readings from before a probe's install date, but nothing rebuilds the profile while the detector runs, so the profile may drift from the creek's rhythm as the seasons change, until the calibration is rerun.

The expected false flags in Table 5 come from the plan's count of 17 sensors, while the shipped calibration judges 30. Across the network, the calibrated stop therefore allows about two false flags a week rather than one, and WaterStrider measured about 1.2.

Finally, every number in this document describes the calibration made on 14 October 2026 and the evaluation run on 25 October 2026. The tails and thresholds are fitted quantities, and they will move when the calibration is rerun.

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